

workers in routine, codifiable tasks while complementing more complex ones. Autor et al. (2003) documented how computerization reduced demand for routine cognitive and manual work, leading to job polarization. Acemoglu and Autor (2011) emphasized that technology replaced mid-skill tasks while raising demand for high-skill labor, and Autor and Dorn (2013) showed that this was accompanied by growth in low-skill service jobs. More recently, Acemoglu and Restrepo (2022) estimated that automation explains a large share of rising U.S. wage inequality since 1980. While this literature focuses on differences across education or occupations, our paper extends the analysis to seniority within firms. We ask whether GenAI is a “*seniority-biased*” technological change, disproportionately affecting juniors who typically perform simpler, more routinized tasks even in high-skill fields.⁸

Evidence from Firm-Level AI Adoption: A related strand of work—closest to our empirical approach—examines the implications of firm-level AI adoption. Babina et al. (2024) construct a measure of firm-level AI investment by combining online résumé data from Cognism with Burning Glass postings. Their results suggest that for U.S. firms in the 2010s, AI-adopting firms grow faster in sales, employment, and innovation, with workforces becoming more educated and technologically oriented. Acemoglu et al. (2022) similarly use Burning Glass postings from 2010–2018 to identify AI-exposed establishments based on tasks and skills in vacancies. They find that exposure is associated with lower hiring at the establishment level, but aggregate occupation/industry effects are too small to detect over that period. More recently, Hampole et al. (2025) use similar data and NLP methods to develop measures of firm-level AI adoption and task-level exposure, which we closely follow in our analysis. They infer adoption from résumé text, using a large language model to extract in-house AI applications and map them to O*NET tasks via sentence embeddings, thereby identifying tasks exposed to AI. They find that between 2010 and 2023, higher exposure corresponds to lower labor demand, but that firms’ productivity gains offset job losses by expanding employment elsewhere, resulting in muted

⁸While it is usually not the main focus, the SBTC literature has also examined variation by age/experience. For instance, Acemoglu and Autor (2011) show that the surge in the college wage premium during the 1980s was concentrated among less experienced workers, whereas from the mid-1990s onward, the increase was driven primarily by more experienced cohorts. They argue this pattern is consistent with changes over time in relative supply by experience and education (and imperfect substitution across experience groups).

net changes in total headcount.⁹

These studies highlight that pre-2023 adoption of AI technology often entailed internal reallocation rather than aggregate job loss. In contrast, our paper provides U.S. evidence on the implications of firm-level adoption during the first years of widespread *Generative* AI diffusion (2023–2025). By focusing not only on overall labor demand but on within-firm seniority composition, we provide evidence that GenAI-adopting firms reduce junior employment while leaving senior employment unaffected.

The Effects of Generative AI on the Labor Market: Finally, a rapidly growing empirical literature examines the labor-market effects of GenAI. Experimental studies generally find that GenAI complements less-experienced workers by boosting their productivity. For example, [Noy and Zhang \(2023\)](#) show that access to ChatGPT substantially reduces completion time and improves output quality, with especially large benefits for lower-ability workers. [Brynjolfsson et al. \(2025b\)](#) similarly find that GenAI assistance in customer support raised productivity by roughly 14 percent on average, with the largest gains for novices. [Dell’Acqua et al. \(2023\)](#) report comparable improvements in consulting workflows. Moreover, [Cui et al. \(2025\)](#) find especially high productivity gains and adoption rates for less experienced software developers. Related field-experimental evidence in [Dell’Acqua et al. \(2025\)](#) shows that adding a GenAI copilot reshapes teamwork and the division of expertise, shifting routine cognitive work to the tool and reorienting human effort toward higher-level tasks. These findings are consistent with the view that GenAI can act as a “leveler,” narrowing productivity gaps between less and more experienced workers ([Autor, 2024](#)).

A second wave of studies uses large-scale labor-market data to track employment dynamics across occupations by GenAI exposure. Closest to our work, [Brynjolfsson et al. \(2025a\)](#) show that since the late-2022 debut of GenAI, employment of young entry-level workers (ages 22-25) in the most AI-exposed occupations fell by about 13 percent relative to trend, while more experienced workers in those occupations saw stable or ris-

⁹Two related studies examine the effects of *Generative* AI adoption. [Humlum and Vestergaard \(2025\)](#) provide evidence from Denmark, linking large-scale worker surveys on GenAI adoption to matched employer-employee data. Despite rapid adoption, they do not find effects of adoption on earnings and hours, suggesting that labor-market impacts in Denmark remain minimal. [Chen and Stratton \(2025\)](#) analyze firm-level adoption of GitHub Copilot and Cursor using detailed engineering workflow data, documenting effects on productivity, task allocation, and organizational collaboration.

ing employment. [Simon \(2025\)](#) documents that entry-level job postings have declined more than 35 percent since January 2023, with the steepest drops in highly exposed roles: a 10-point increase in exposure predicts an 11 percent decline in entry-level demand, while senior roles in those same occupations increase by 7 percent. [Dominski and Lee \(2025\)](#) link occupational exposure scores to CPS data and, using a first-difference design, show that higher GenAI exposure is associated with reduced employment. More recently, [Lodefalk et al. \(2026\)](#) use Swedish job-posting and register data in an employer-level difference-in-differences design and find a widening age gradient in hiring: since the release of ChatGPT, employment of workers aged 22–25 in high-exposure occupations has fallen by about 5.5 percent relative to less-exposed occupations within the same employers, while workers over 50 have seen a slight increase. [Crane and Soto \(2026\)](#) focus on computer-programming occupations—among the most AI-exposed—and, using a within- versus between-industry decomposition of CPS data, find that coder employment growth is roughly 3 percentage points lower post-ChatGPT than pre-ChatGPT, even after controlling for industry-level shocks.¹⁰ By contrast, [Chandar \(2025\)](#) and [Financial Times \(2025\)](#) do not find systematic differences in employment patterns between more- and less-exposed occupations in CPS data. [Eckhardt and Goldschlag \(2025\)](#) compare unemployment patterns using five exposure measures, finding statistically significant differences for only two, with even those effects relatively small.

Our contribution to this literature is to move beyond occupation-level exposure indices and provide broad-based evidence using a more direct measure of firm-level adoption, identified through postings for explicit “GenAI integrator” roles. Moreover, unlike exposure-based measures, our adoption measure varies over time, allowing us to exploit variation in adoption timing with an event-study design.

¹⁰Motivated by this body of evidence, [Friebel et al. \(2026\)](#) model the dynamic implications of reduced junior hiring for the short- and long-run structure of firms’ workforce. Relatedly, [Ide \(2025\)](#) develop an overlapping-generations model in which AI-driven entry-level automation weakens the intergenerational transmission of tacit knowledge from experts to novices, potentially reducing long-run growth.

3 Conceptual Framework

3.1 Overview

This section develops a simple formal framework to analyze the “seniority-biased” labor demand effects of GenAI. The objective is to explain why GenAI adoption may disproportionately reduce demand for junior workers and to clarify potential mechanisms underlying this pattern.¹¹ The framework is motivated by two broad patterns in the emerging evidence: micro-level experimental studies consistently find that GenAI disproportionately increases the productivity of less-experienced workers (Brynjolfsson et al., 2025b; Noy and Zhang, 2023; Dell’Acqua et al., 2023; Cui et al., 2025), whereas labor market evidence, including our results, suggests that GenAI adoption and exposure are associated with declines in junior employment (Brynjolfsson et al., 2025a; Azar et al., 2025; Klein Teeselink, 2025; Lodefalk et al., 2026).

The model examines how GenAI adoption reshapes the composition of labor demand between junior and senior workers in a partial-equilibrium setting. It highlights three mechanisms: a *displacement effect*, whereby automation removes tasks previously performed by juniors; a *productivity effect*, whereby higher productivity reduces labor required on surviving tasks; and a *reallocation effect*, whereby juniors expand into some previously senior tasks as their relative productivity rises. The main result is that when tasks are complementary or Cobb-Douglas ($\sigma \leq 1$), displacement and productivity effects reduce junior labor demand, while reallocation is the only offsetting force. Whether junior employment declines more than senior employment therefore depends on the relative strength of these forces.

3.2 Setup

We build on the task-based framework of Acemoglu and Autor (2011) and Acemoglu and Restrepo (2018). We consider a short-run partial-equilibrium setting in which firm output Y and wages are held fixed. The goal is to isolate how adoption of GenAI by a firm

¹¹This framework explains the *incidence* of GenAI across seniority levels—why juniors may bear the labor demand cost while seniors do not. It is distinct from, and complementary to, the forward-looking hiring model in Supplemental Appendix A.5, which explains the *timing* of the employment response—why firms adjust junior hiring early, before automation is fully realized.